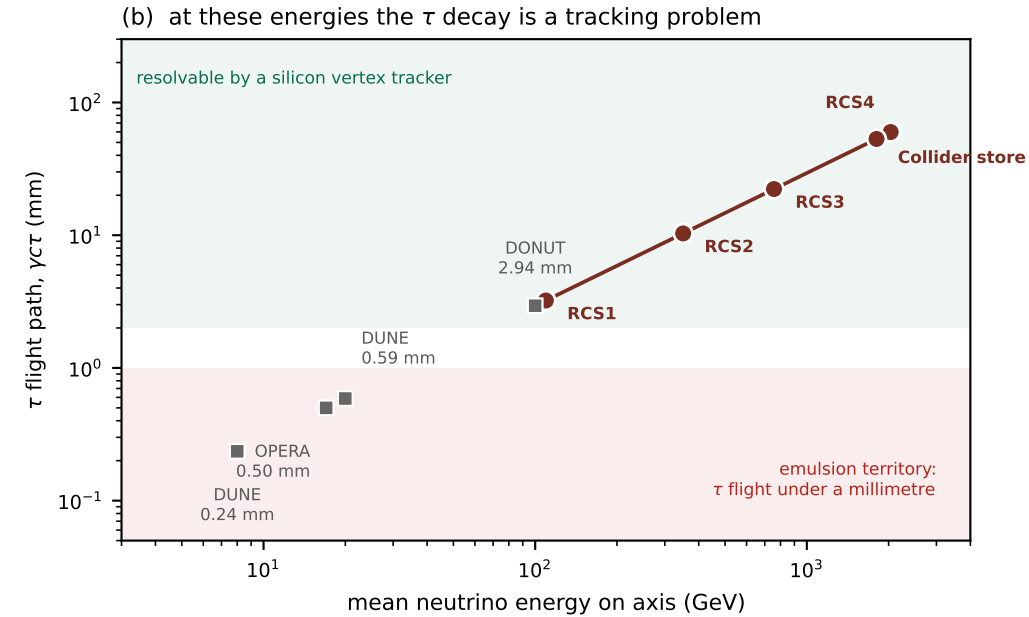
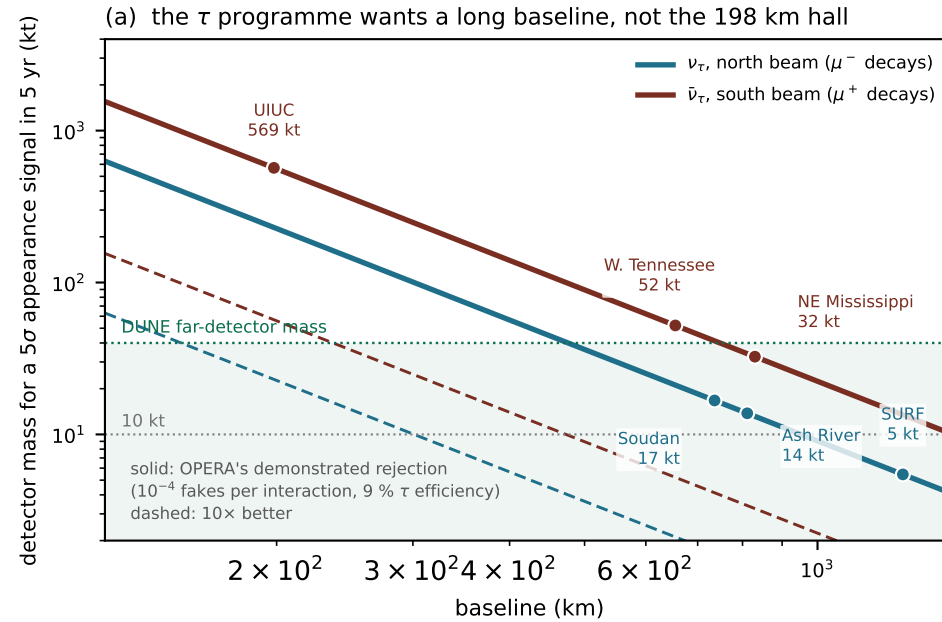
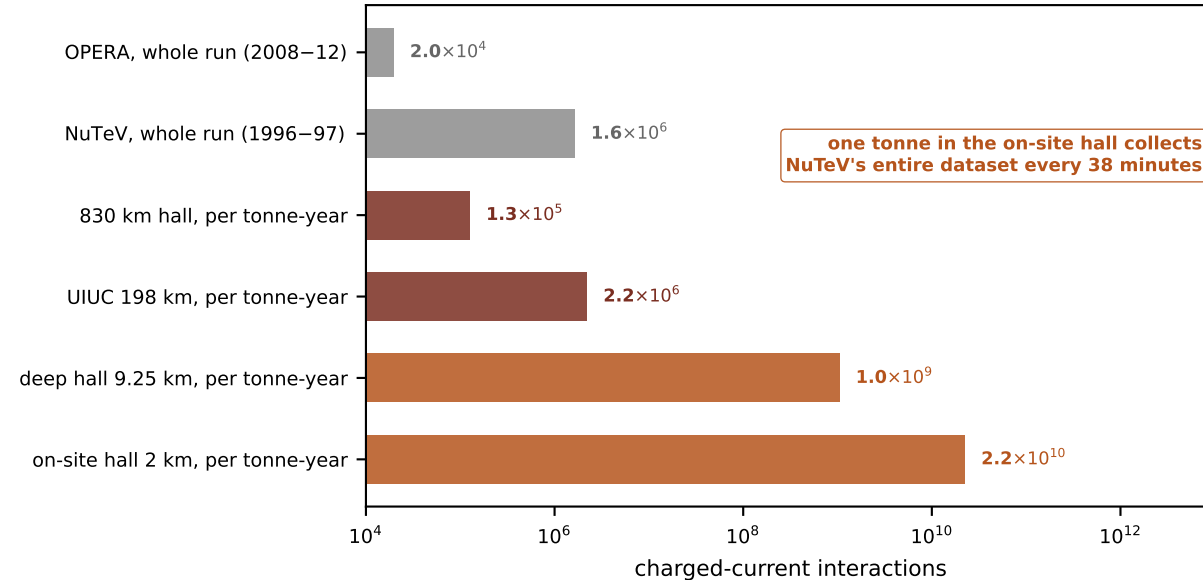


# What the corridor beam is worth

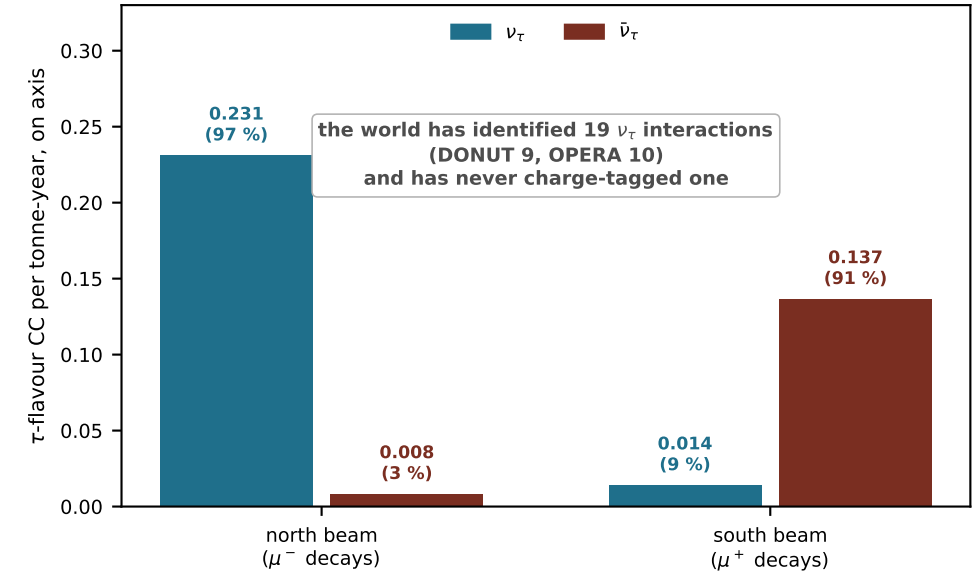
$\mu^+$  and  $\mu^-$  counter-rotate, so one straight fires a sign-tagged  $\bar{\nu}_\tau$  beam south and a  $\nu_\tau$  beam north, from a flux that is pure muon decay



(c) two whole experiments, and one tonne-year of this beam



(d) one straight, two sign-tagged beams, fired at once



Rates from tools/tau\_compass.py and tools/nutau\_sites.py (on-axis point detector, CSMS cross-sections,  $\tau$ -threshold suppression). Reach uses the chirp as a weighted fit,

$Z = \varepsilon \sqrt{(MT/f) \sum_i S_i^2 / B_i}$ , with OPERA's own  $\tau$  efficiency and fake rate. Comparison numbers: DONUT PRD 78 052002, OPERA PRL 120 211801, NuTeV PRL 88 091802, DUNE TDR arXiv:2002.03005.